Smart Home Controller System

Home Owner

- Enter Login Credentials
- Provide Biometric Info
- Select Option
- Access Reports Menu
- Send Command to Smart Device
- Create Automation Schedule
- Register New Device
- Log Out
- Select Report Type
- Select Smart Device
- Enter Smart Device Info
- Select Command
- Enter Schedule Information
- Save Info of New Smart Device
- Connect New Smart Device to Network
- Sync and Connect with New Smart Device
- Display Registration Successful
- Display Bye!

Smart Home Controller

- Authenticate Login Credentials
- Are login credentials correct?
  - Yes
  - No
- Request Biometric Info
- Display Auth. Error
- Verify Biometric Info
- Is Biometric Valid?
  - No
  - Yes
- Deny Access
- Grant Acess
- Display Menu Options
- Check for Software Update
- Update Available?
  - Yes
  - No
- Download & Install
- Restart System
- Get Energy Data from Smart Device
- Calculate Energy Usage
- Determine Energy Optimizaton Route
- Generate Energy Log
- Generate Energy Optimization Report
- Save Energy Log
- Save Energy Optimization Report
- Listen for Voice Command
- Command recognized?
  - Yes
  - No
- Execute Command
- Prompt User to repeat
- Request Data from Security Senors
- Analyze Data
- Is there suspicious activity?
  - No
  - Yes
- Send Alert to User's mobile device
- Security Report
- Send Security Report to User's mobile device
- Energy Optimization Report
- Send Energy Optimization Report to User's mobile device
- Process Command
- Send Command Signal to Smart Device
- Process Schedule Info
- Implement Schedule for Smart Device

Smart Device

- Send Data
- Receives Command Signal
- Execute Command